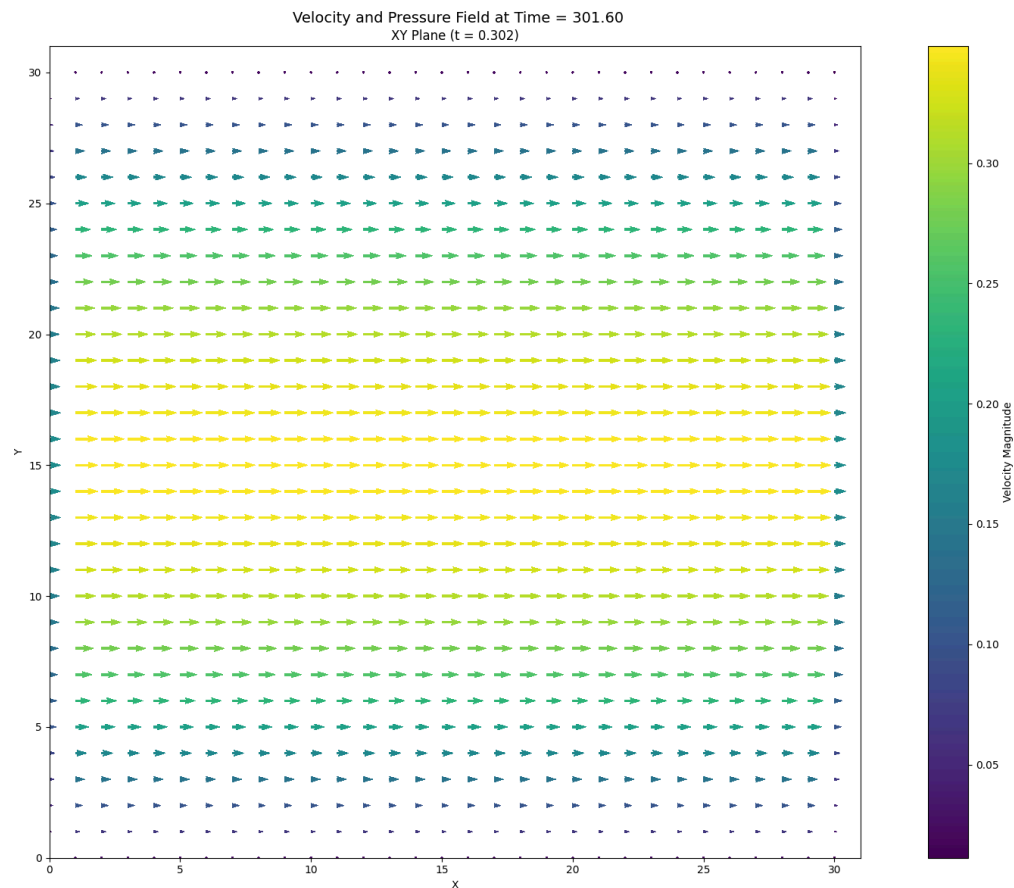


Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	9.20000000e+02	kg m ⁻³
Dynamic viscosity	ν	8.10000000e-02	Pa s
Kinematic viscosity	ν	8.80434783e-05	m ² s ⁻¹
Characteristic velocity	u_{max}	1.00000000e+00	m s ⁻¹
Characteristic length	D	5.00000000e-03	m
Reynolds number	Re	1.97740055e+01	dimensionless
Body force	G	9.81000000e+00	m s ⁻²
Grid points	N_x	3.10000000e+01	points
Unused transverse grid points	N_y	3.10000000e+01	points
Grid spacing	dy	1.61290323e-04	m

Quantity	Symbol	Value	Unit
Time step	dt	1.31646880e-04	s
Numerical threshold	epsilon	1.00000000e-15	-

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
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Parameters listed above are the values used by the selected solver.

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, y)	(2143, 31)
Regression field shape (t, y)	(543, 27)
Regression samples	14,661
Dictionary terms	7
First saved time index	150
Temporal crop at each boundary	800 layers
x-boundary crop	0 points
y-boundary crop	2 points

4. Governing Equation Verification

Reference equation

$$u_t = 8.80434783e-05 * u_{yy} + 9.81000000e+00 * one$$

Learned equation

$$u_t = 8.84726274e-05 * u_{yy} + 9.85781675e+00 * one$$

Term	Reference coefficient	Learned coefficient	Absolute error
$u * u_y$	+0.00000000e+00	+0.00000000e+00	0.00000000e+00
$u * u_{yy}$	+0.00000000e+00	+0.00000000e+00	0.00000000e+00
$u_y * u_{yy}$	+0.00000000e+00	+0.00000000e+00	0.00000000e+00
u	+0.00000000e+00	+0.00000000e+00	0.00000000e+00
u_y	+0.00000000e+00	+0.00000000e+00	0.00000000e+00
u_{yy}	+8.80434783e-05	+8.84726274e-05	4.29149147e-07

Term	Reference coefficient	Learned coefficient	Absolute error
1	+9.81000000e+00	+9.85781675e+00	4.78167517e-02

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	1.00000000e-06
L0 penalty	1.00000000e-06
Training RMSE	1.11757497e-13
Validation RMSE	1.24716034e-13

6. Reproducibility Notes

- Input directory:
/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_poiseuille/u_velocity_field
- Fields were loaded starting at saved time index 150.
- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the supplied reference coefficients.